Exploration of Person-Specific Functional Organization in Right Prefrontal Cortex

by

Yiran Fan

Faculty Mentor: Thad Polk, Ph.D.

Graduate Student Mentor: Katy Michon

Reader: Daniel Weissman, Ph.D.

University of Michigan

April 1, 2023

A thesis submitted in partial fulfillment

of the Degree of Bachelor of Science

in Cognitive Science with Honors

Abstract

Cognitive control is an essential brain function, but how this function is implemented in the brain is not yet well understood. One reason for this lack of clarity may be the use of group-averaged results in traditional neuroimaging approaches. However, recent studies using precision neuroimaging have explored the details of the brain's neural architecture, revealing person-specific details that were missed using a group-averaging approach. In this study, we investigated whether the functional organization of right prefrontal cortex (PFC) is person-specific and reliable within a person. We repeatedly scanned two individuals for approximately 9 hours each while they completed an unusually large task battery spanning multiple different areas of cognitive control. We measured the within- and between-subject similarity of patterns of activation from these two individuals and found that across nearly 88 contrasts, within-subject similarity was higher than between-subject similarity. An exploratory factor analysis revealed eight latent factors underlying activation differences across right PFC vertices which we tentatively associate with performing a task in general and task contexts. These results suggest that the functional organization of cognitive control is person-specific, or unique to a given individual, but also reliable within a given individual. Furthermore, these results provide support for the utility of precision neuroimaging approaches in furthering our understanding of the brain's functional organization.

Keywords: precision neuroimaging, fMRI, exploratory factor analysis, cognitive control, functional organization, right prefrontal cortex

Scientific Acknowledgement

The experiments presented in this thesis resulted from a close collaboration with Katy Michon, a graduate student in Dr. Thad Polk's Computational and Cognitive Neuroscience Lab, and she has granted me permission to include in this thesis some of the data she collected and analyzed.

Introduction

Cognitive control is the process that allows goal-directed behaviors (Miller & Cohen, 2001). In daily life, many behaviors require cognitive control, such as maintaining a shopping list in mind when going to the grocery store and resisting the impulse to browse social media when studying for an exam. As an essential brain function, cognitive control reduces uncertainty in decision-making by prioritizing goals and motivations (Dixon & Christoff, 2012; Gabrys et al., 2018). It plays an important role when there is competition for limited mental resources in the brain (Desimone & Duncan, 1995). Yet despite numerous neuroimaging studies of cognitive control, how this function is implemented in the brain is not well understood.

The prefrontal cortex (PFC) in particular has long been associated with cognitive control in the brain. However, compared to its left-hemisphere counterpart, less is known about how the right PFC contributes to cognitive control. Previous research suggests that the right dorsolateral, ventral lateral, and orbitofrontal PFC may be related to action updating and working memory (Barbey et al., 2013; Levy & Wagner, 2011; Wikenheiser & Schoenbaum, 2016), while other studies indicate that these areas are also associated with inhibitory control (Aron et al., 2014; Garavan et al., 1999; Suda et al., 2020). Such studies often focus primarily on understanding the role of localized brain areas in a specialized cognitive process. Nevertheless, the right PFC possesses complex functional networks where cognitive functions are usually not implemented by isolated brain regions (Kane & Engle, 2002; MacDonald et al., 2000; Menon & D'Esposito, 2022; Ridderinkhof et al., 2004). Thus, how cognitive functions are subserved by the right PFC in general is not yet well understood.

One reason for this lack of clarity may be the use of group averaging in neuroimaging. Traditionally, many neuroimaging studies collect a small amount of data (typically only 5-10

minutes; Gordon et al., 2017b; Laumann et al., 2015) from a large number of people, and average patterns of activation across the group. Under this framework, the individual's unique patterns of brain activity are usually considered noise (Van Horn et al., 2008). This approach is partially based on the fact that all brains share many features. However, previous research has shown that the brain's anatomical and functional organization vary across individuals (Fedorenko & Kanwisher, 2009; Gordon et al., 2017a; Juch et al., 2005; Laumann et al., 2015; Tomaiuolo et al., 1999) Thus, averaging brain activity across functionally distinct brain regions might obscure details that could be used to better understand brain function (Gordon et al., 2017b; Laumann et al., 2015).

Recent studies have taken an alternative approach, called "precision" neuroimaging, where fewer people are scanned for multiple hours (often more than 2) to obtain more detailed and reliable data than is possible with traditional approaches (Gratton et al., 2020; Smith et al., 2021). This method addresses some limitations associated with group-averaged results by producing more reliable within-person measurements and accounting for individual differences in brain structure (Michon et al., 2022). Utilizing this approach, some precision neuroimaging studies have shown that the functional organization of individual brains, especially in areas related to cognitive control, is highly variable (Laumann et al., 2015; Seitzman et al., 2019; Smith et al., 2021). Braga and Buckner (2017) investigated the network organization of four individuals with 24 hours of MRI each and discovered that the default network previously studied by the group-averaging approach might be comprised of several different adjacent networks in different brains. Similar studies have also focused on specific regions related to cognitive control. For example, a study by Seitzman et al. (2019) has shown that the lateral frontal and parietal cortices exhibit distinct between-subject functional connectivity that may be

associated with higher-order cognitive processes using three precision neuroimaging datasets. Functional connectivity profiles, especially in the frontoparietal network, are also specific to individuals (Finn et al., 2015; Mueller et al., 2013). Multiple studies have also demonstrated that individual brains often differ from group-averaged results. In a review by Miller and Van Horn (2007), they suggested that patterns of brain activity associated with episodic memory retrieval were unique and not always reflected by those at the group level. Similarly, Gordon et al. (2017b) studied the "Midnight Scan Club" dataset, a precision neuroimaging dataset consisting of over ten hours of resting and task-related fMRI data for each participant, where they found reliable person-specific features in brain activations for motor and memory-related tasks that were not presented in group averages. Such studies shed light on our understanding of cognitive control by mapping the neural heterogeneity across subjects (Smith et al., 2021). With the current insights, precision neuroimaging becomes a promising path to study cognitive control and its underlying functional neuroanatomy.

In its earliest days, most precision neuroimaging research focused on resting-state fMRI (RS-fMRI), where data are collected when subjects lie in the scanner without being given a specific task (Dubois & Adolphs, 2016). However, incorporating tasks into scanning provides information related to cognitive function, and is becoming more common. This shift is partially due to the unconstrained nature of the resting state, which potentially induces more noise (Finn et al., 2017). Some studies suggest that individual differences in brain activity and functional parcellations are dependent on the brain state and even change with the task state (Geerligs et al., 2015; Salehi et al., 2020). Although others have found little difference between task-related brain states and resting state, task-based precision neuroimaging may still provide benefits over the resting state (Braga et al., 2020; Gratton et al., 2018). Additionally, utilizing task-related

precision neuroimaging may require less than 25 minutes of data to acquire reliable differences between individuals (Anderson et al., 2011; Finn et al., 2017). For example, Anderson et al. (2011) found that the functional connectivity acquired from resting state and viewing videos are different for the same individual, but both are person specific. The authors also suggested that task-related functional connectivity requires less imaging time to produce reliable and accurate results when predicting individual differences. Furthermore, studying the brain in multiple contexts may allow a better understanding of the functional architecture underlying individual differences (Geerligs et al., 2015; Pinho et al., 2021). Several studies examined the Individual Brain Charting Project (Pinho et al., 2018), an MRI dataset where participants completed approximately 30 tasks, and suggested that an increasing number of tasks and the corresponding functional contrasts during imaging help to distinguish functional regions of cognitive control and produce a finer-grained details (Pinho et al., 2021; Thirion et al., 2021). Thus, the combination of task batteries and precision neuroimaging may provide further insight into individual functional organization than is possible with RS-fMRI alone.

In line with these ideas, we have combined precision neuroimaging with an unusually large task battery to better understand the functional organization of cognitive control. This study aimed to investigate whether the functional organization of the right PFC is person-specific, or unique to a given individual, and reliable within a person using a precision neuroimaging dataset collected at the Computational and Cognitive Neuroscience Lab at the University of Michigan. Specifically, we measured within- and between-person similarity of patterns of activation taken from 88 functional contrasts from two separate individuals. Because we believe that the functional organization of the right PFC is person-specific and reliable, we hypothesized that within-person similarity would be greater than between-person similarity. In other words, we

hypothesized that patterns of activation taken from independent datasets from a single individual would be more similar to each other than to patterns of activation from another person. We further conducted an exploratory factor analysis (EFA) to examine latent factors that explain the main sources of variation in brain activity across tasks and vertices.

Methods

Participants

Two participants (both female, aged 32 and 25) were included in the dataset. Both participants were right-handed and declared no contraindications for the functional magnetic resonance imaging (fMRI) procedure. Both participants were given written informed consent. The study was approved by both the University of Michigan Health Sciences and Behavioral Sciences Institutional Review Board (IRB-HSBS; for experimental tasks) and the Institutional Review Board of the University of Michigan Medical School (IRBMED; for MR procedures).

Image Acquisition

The fMRI data were acquired using a GE MR750 3T MRI scanner (General Electric, Waukesha, WI) at the University of Michigan fMRI Laboratory. Head padding with foam was used to stabilize the head in all sessions. The button responses were collected with Celeritas 5-button fiber optic response system (Psychology Software Tools, Sharpsburg, PA). Each scan was between 75 and 90 minutes. For each scan, T1-weighted anatomical images, high-resolution anatomical images using spoiled 3D gradient-echo acquisition (SPGR), and T2*-weighted functional images with single-shot gradient-echo (GRE) reverse spiral pulse sequence were collected. The following parameters were used in fMRI data collection: field of view=220x220mm, voxel size=3x3x4mm, 40 axial slices, TR=2000ms, TE=30ms.

Task Design and Procedure

Each participant completed approximately 9 hours of fMRI scanning, including 18 different cognitive tasks. They completed each task twice over the course of six sessions in a test-retest design (Table 1). The tasks were administered in the same order for each pair of test-retest sessions. Before each task session, the participants practiced each task outside of the scanner to familiarize themselves with the task instructions. Experimental tasks were presented using PsychoPy (Peirce et al., 2019). The tasks were adapted from previous publications of fMRI studies exploring some aspect of cognitive control. All tasks were adapted to be approximately 6-8 minutes per run, and participants typically completed 1 run of each task per session (except in sessions 3 and 4; see Table 1).

Table 1

Task Design and Descriptions

Task	Descriptions
Color Shape Task (CS; Hakun et al., 2015)	Stimuli had 2 possible shapes (circle or square) and two possible colors (red or blue). The participants were shown a cue and instructed to respond to either the shape of the stimuli or the color of the stimuli or switch between the two. A block cue lasting 1.5s was presented before each block. There were 20 trials in each block, consisting of a 0.15s trial cue and a 2.65s stimulus with 0.2s inter-trial-interval (ITI). The task included 8 color/shape-only blocks (4 of each) and 5 task-switching blocks with 10s fixation in between blocks.
Number Letter Task (NLE; Reineberg et al., 2018)	Stimuli consisted of number-letter pairs that were shown in one quadrant of the screen. Participants were instructed to indicate whether the number was even or odd if the pair was in a top quadrant and whether the letter was a vowel or a consonant if the pair was in a bottom quadrant. In each trial, the quadrant where the stimuli would be shown was highlighted for 0.35s, with a 2.3s stimulus and 0.35s ITI. The task was organized into 7 trials per block and 11 blocks in total. The first 3 blocks consisted of pairs shown only in the upper quadrants; the next 3 blocks included pairs shown only in the bottom quadrants; and the last 5 blocks were shifting blocks where pairs could appear in any quadrant.
Number List Task (NLI; DiGirolamo et al., 2001)	Participants were shown a list of repeating numbers. In the first 3 blocks, participants indicated whether the number composing the list was greater than 5. In the next 3 blocks, participants indicated whether the list length was greater than five digits. In the final 6 blocks, participants alternated between value and length judgments. Each trial began with a 0.15s cue, followed by a 2.85s stimulus and a 0.25s ITI.

Table 1 Continued

Task	Descriptions
Grid Updating Task (GU; Leung et al., 2007)	Participants were instructed to remember the position of a red circle on a 4x4 grid and mentally update the dot's position based on a series of 12 arrows or dashes that indicated the direction of movement around the grid. If a dash was presented, there was no update. The original dot was presented for 1s, and the updates were shown for 0.25s each with a 0.45s interval. At the end of the trial, participants responded to whether a probe matched the dot's new position. The probe was shown for 1s, followed by a 1.5s ITI. Each block contained 2 trials. The task was organized into 10 alternating updating blocks and control blocks (blocks with only dashes presented) with 10s resting fixation in-between.
Letter N-Back Task (NBACK; Demetriou et al., 2018)	The task consisted of 0-back blocks, in which the participants were instructed to indicate whether the current letter was X, and 2-back blocks, in which they were instructed to indicate whether the current letter was the same as two trials before. There were 10 trials in each block, and each trial lasted 1.5s with 0.5s ITI. Twelve alternating 0-back and 2-back blocks were administered with 10s fixations in between each.
Sternberg Task (STE; Altamura et al., 2007)	In the test condition, participants were presented with a string of letters followed by a single probe letter and indicated whether the probe was in the string. In the control condition, they were shown a string of arrows followed by a probe arrow and responded to the direction of the probe arrow. Each stimulus appeared for 2s, followed by a 1s delay and a 1s probe. The ITIs were 1s. The task blocks were modified to 4 trials per block with 6 test blocks and 5 control blocks with 10s fixations in between each.
Stroop Task (STR; Kikuchi et al., 2012)	In the color identification blocks, the participants were shown the text "XXXX" in either red, blue, or green font and instructed to respond to the color. In the incongruent blocks, they were shown color names (e.g., "RED") with an incongruent font color and instructed to indicate the font color. Each stimulus was presented for 1.9s, followed by a 0.1s ITI. There were 12 alternating color identification and incongruent blocks with 10 trials per block.
Simon Task (SIM; Georgiou-Karistianis et al., 2007)	Participants were shown arrows pointing to either direction on the left or right side of the screen. The experimental blocks consisted of congruent trials (arrow with matching direction and screen location; e.g., left/left) and incongruent trials (arrow with a non-matching direction/screen location). Participants were instructed to respond to the direction the arrows were facing. The baseline blocks included only horizontal lines and involved no response. Task blocks were modified to 12 alternating experimental blocks and control blocks with 6 trials per experimental block and 4 trials per baseline block. The trials each lasted 2.5s with 0.5s ITIs.
Flanker Task (FLA; Wager et al., 2005)	Stimuli were three circles in either red, green, blue, or yellow. Participants were instructed to respond to the color of the target (central circle) with the left hand if it was red or green and with the right hand if it was blue or yellow. The flanking circles were always the same color, but different from the target. There were 12 alternating congruent blocks (target and flankers indicated the same response) and incongruent trials (target and flankers indicated different responses) with 10s fixation in between. Each block had 12 trials with 1s of stimuli and 0.44 ITI.
Inhibit, Switch, and Update Task (ISU; Lemire-Rodger et al., 2019)	This task used an event-related design. Stimuli were three yellow and/or blue squares. During the control condition, participants were instructed to indicate whether the central square was blue. During the updating condition, they were instructed to indicate whether the central square was blue two trials ago. And in the inhibition condition, they were instructed to indicate the same thing as in the control condition, unless two squares were yellow. The task consisted of 5, 9-block runs. There was a 30s resting baseline before and after each run. Each block began with a 1s cue and a jittered interval of 1-5s. During the block, each stimulus was presented for 0.5s with a jittered ITI of 2-6s. There were 12 trials per block.

Table 1 Continued

Task	Descriptions
Counter Switch Task (COUNTSW; Sylvester et al., 2003)	Participants were shown a series of left and right arrows and instructed to keep track of how many they'd seen of each. Then, they were asked to indicate how many left or right arrows they saw. There were 12 alternating high-switch blocks (consisting of 70% switching trials) and low switch-block (consisting of 20% switching trials) with 10s fixations in between. Eleven arrows were presented per block and appeared for 1s followed by a 0.44s ITI. At the end of each block, the interval before the last arrow and the probe was 0.6s. The probe was presented for 4s.
Stimulus-Response Compatibility Task (SRC; Sylvester et al., 2003)	Participants viewed a pseudorandom list of left- and right-facing arrows. During compatible blocks, participants responded to the arrows using the same hand indicated by the arrow (e.g., left hand for left arrow). During incompatible blocks, participants responded with the opposite hand (e.g., right hand for left arrow). A total of 12 alternating compatible and incompatible blocks with 10s fixations in between were administered. Each block began with a 1.5s cue and included 11 trials, each 1s long with a 0.44s ITI.
Spatial Working Memory Task (SPAWM; Silk et al., 2010)	Participants were shown either five black squares (high-load condition) or two black squares (low-load condition) and instructed to remember their locations. After a 5s retention period, a single black square was shown, and the participants indicated whether this square was in the same location as any of the original stimuli. Stimuli lasted for 3.85s and the probe was shown for 1s after the retention period. There were 8 blocks of alternating high and low-load blocks, each with 3 trials and a 0.15s ITI.
Overt Attention Task (OVERT; de Haan et al., 2008)	Stimuli consisted of a central circle with an overlaid white cross surrounded by 8 other circles. In the peripheral condition, the arm of the central cross was red and rotated to point to one of the surrounding circles. If the circle the cross pointed to was smaller than the others, participants were instructed to make a rapid saccade to that circle. In the central condition, the participants were instructed to maintain fixation on the central cross. There were 12 alternating central and peripheral blocks with 10s fixations in between. Each block had 20 trials and lasted 20s. The circles were visible for 0.4s each with a 0.6s ITI.
Covert Attention Task (COVERT; de Haan et al., 2008)	The visual stimuli, timing, and block design were similar to those of the Overt Attention Task. In this task, participants were instructed to maintain fixation on the central circle/cross at all times. In the peripheral condition, participants made a button response if the cross pointed to a smaller circle. In the central condition, participants made a response if the central circle became smaller than normal.
Random Generation Task (RNG; Gilbert et al., 2008)	Participants were shown a timing signal (a flashing square) during all blocks and instructed to press one of four buttons as the signal appeared. In the experimental conditions, they pressed the buttons in a random order, while in the baseline conditions, they pressed the response buttons in a stereotypical left-to-right sequence. There were 30 signals in each block, each lasting 0.7s with a 0.3s interval in between. Each condition was presented alternatingly 6 times with 10s fixations in between.
Go No-Go Task (GNG; Wager et al., 2005)	This task was block-designed but analyzed as event-related. Participants saw a random series of letters on the screen, each lasting 0.44s, followed by a 1s ITI. They were instructed to respond to each letter unless it was an "X". The task contained high (50%) and low (20%) no-go trial blocks. Each block contained 12 trials. Ten alternating high no-go and low no-go blocks were administered with no resting fixations in between.
Item Recognition Task (ITEMREC; Awh et al., 1996)	In the experimental condition, four upper-case letters appeared for 0.2s, and a lower-case letter probe followed after a 3s delay. Participants were instructed to indicate whether the probe was one of the previous letters. In the control condition, similar stimuli were presented, but stimuli were delayed 3s initially and then remained on the screen as the probe appeared. There were 4 trials in each block and 12 alternating experimental and control blocks with 10s fixations in between.

Note. Descriptions of tasks completed during the fMRI scanning sessions. The first 9 tasks were completed in session 1 and again in session 2. In sessions 3 and 4, they completed the inhibit, switch, update task, and in sessions 5 and 6, they completed the final 8 tasks.

fMRI Pre-processing

The fMRI data were preprocessed using Freesurfer's functional analysis stream (FS-FAST; Fischl, 2012). Preprocessing included constructing two-dimensional cortical surfaces from three-dimensional volumes collected during the scans, resampling them to *fsaverage* space (163,842 vertices per hemisphere), motion correction using the first volume of the first run as a reference, slice-timing correction, and smoothing using a 5mm Gaussian kernel. A set of 88 contrasts were then computed in the first-level data analyses, all of which were used in the final analysis dataset.

Data Analysis

The task runs were randomly assigned to either a "Set1" or "Set2" dataset to reduce the effect of different scanning sessions for each participant. The contrasts from these two datasets were then organized into two "Set1" and "Set2" matrices for each subject, resulting in four 163,842 x 88 contrast matrices (person 1, set 1; P1S1; person 1, set 2; P1S2; person 2, set 1; P2S1; person 2, set 2; P2S2). The rows of each matrix were the vertices of the right hemisphere (163,842), while the columns of each matrix were the contrasts computed during the first-level analyses (88). The Desikan-Killany atlas (Desikan et al., 2006) was used to identify vertices representing right prefrontal cortex (PFC). Ten regions were selected for analysis based on their association with cognitive control and with right PFC in previous literature: superior frontal gyrus, rostral and caudal middle frontal gyri, inferior frontal gyrus (pars opercularis, pars triangularis, and pars orbitalis), lateral and medial orbitofrontal gyri and rostral and caudal anterior cingulate gyri (Aron & Poldrack, 2006; Banich et al., 2000; Barbey et al., 2013; Garavan et al., 1999; Wikenheiser & Schoenbaum, 2016). We restricted our analyses to an

anatomical mask of these regions by removing all other rows (vertices) from the matrices. This process resulted in four 38,652 x 88 (vertex x contrast) matrices in the final analyses.

To measure within- and between-subject similarity of the contrast matrices, we first computed Pearson correlations between every contrast for each within- and between-subject pair. Within-subject pairs included correlations from each person across "Set1" and "Set2" matrices (i.e., P1S1-P1S2 and P2S1-P2S2), and between-subject pairs included correlations from either "Set1" or "Set2" matrices across the two participants (i.e., P1S1-P2S1 and P1S2-P2S2). We then averaged the two within-subject correlations and the two between-subject correlations for each contrast and conducted a paired-sample *t*-test comparing the average within-subject correlation with the average between-subject correlation across contrasts to determine whether within-subject similarity was significantly greater than between-subject similarity.

To examine latent factors that explain variation in brain activity, we also conducted an exploratory factor analysis (EFA) in MATLAB (using *factoran*). First, we *z*-scored the four 38,652 x 88 (vertex x contrast) matrices individually and concatenated them into one 154,608 x 88 matrix including both participants and sessions (P1S1, P1S2, P2S1, and P2S2). We chose the number of extracted factors based on the number suggested by the scree plot, as well as those above and below the chosen number. We used an orthogonal varimax rotation for all factor solutions. Finally, we chose an eight-factor solution in which each contrast had high loadings on a single factor and where few contrasts failed to load on any factor.

Results

Within- and Between-Subject Similarity

Results revealed that the within-subject similarity of the contrast data was significantly greater than the between-subject similarity of the contrast data ($t(87) = 8.859$, $p < 0.001$). The within- and between-subject correlations are shown in Table 2. This result can also be observed in brain maps obtained from the contrast data (Figure 1), where patterns of activation from a given contrast in the same person appear more visually similar than those from different subjects. These results suggest that the functional organization of cognitive control is person-specific and reliable.

Table 2

Within- and Between-Subject Correlations of Contrasts

Tasks	Contrasts	Within-Subject Correlation		Between-Subject Correlation	
		P1S1-P1S2	P2S1-P2S2	P1S1-P2S1	P1S2-P2S2
CS	colorshape-vs-fix	0.071	0.148	0.044	-0.010
	color-vs-fix	-0.005	0.030	-0.017	-0.088
	shape-vs-fix	0.041	0.398	-0.124	0.043
	shift-vs-color	0.021	-0.029	0.070	-0.111
	shift-vs-colorshape	0.105	-0.226	0.211	-0.124
	shift-vs-fix	0.260	0.240	0.140	0.030
	shift-vs-shape	0.076	-0.172	0.137	-0.037
NLE	letter-vs-fix	0.191	0.134	0.146	0.210
	numberletter-vs-fix	0.032	0.172	0.103	0.122
	number-vs-fix	-0.031	0.203	0.010	-0.057
	shift-vs-fix	0.390	0.449	0.188	0.266
	shift-vs-letter	0.063	0.219	0.228	0.107
	shift-vs-number	0.217	0.286	0.057	0.062
	shift-vs-numberletter	0.231	0.282	0.127	0.075
NLI	length-vs-fix	-0.026	0.212	0.185	0.129
	numberlist-vs-fix	-0.160	0.100	0.046	0.056
	shift-vs-fix	0.385	0.608	0.249	0.205
	shift-vs-length	-0.071	-0.014	0.107	-0.040
	shift-vs-numberlist	0.044	0.144	-0.070	0.006
	shift-vs-value	0.017	0.281	-0.186	0.092
	value-vs-fix	-0.249	0.154	-0.167	0.037

Table 2 Continued

Tasks	Contrasts	Within-Subject Correlation		Between-Subject Correlation	
		P1S1-P1S2	P2S1-P2S2	P1S1-P2S1	P1S2-P2S2
GU	control-vs-fix	0.217	0.193	0.035	0.158
	fix-vs-gridcontrol	0.640	0.265	0.082	0.442
	update-vs-control	0.201	0.067	0.152	0.175
	update-vs-fix	0.516	0.253	0.202	0.389
NBACK	0back-vs-fix	0.651	0.321	0.123	-0.010
	2back-vs-0back	0.220	0.492	0.186	0.066
	2back-vs-fix	0.687	0.495	0.277	0.368
	fix-vs-2back0back	0.728	0.458	0.263	0.422
STE	control-vs-fix	-0.086	0.218	0.210	-0.133
	fix-vs-sternbergcontrol	0.287	0.382	0.322	0.111
	sternberg-vs-control	0.429	0.227	0.169	0.109
	sternberg-vs-fix	0.609	0.368	0.312	0.270
STR	congruent-vs-fix	0.341	0.037	0.158	-0.027
	fix-vs-inconcon	0.397	0.111	0.180	-0.105
	incongruent-vs-congruent	0.126	0.326	0.161	0.193
	incongruent-vs-fix	0.310	0.245	0.207	0.003
SIM	baseline-vs-fix	-0.095	0.049	0.126	-0.150
	fix-vs-simonbaseline	-0.102	0.137	-0.061	-0.168
	simon-vs-baseline	0.257	-0.067	0.173	-0.003
	simon-vs-fix	0.390	0.083	-0.115	0.006
FLA	congruent-vs-fix	0.263	0.021	-0.106	0.186
	fix-vs-inconcon	0.364	0.121	-0.199	0.323
	incongruent-vs-congruent	-0.108	0.378	0.144	0.085
	incongruent-vs-fix	0.365	0.234	-0.152	0.362
ISU	execfx-vs-control	0.140	0.134	0.022	0.290
	execfx-vs-fix	0.523	0.578	0.258	0.338
	inhibit-vs-control	0.026	-0.041	0.106	0.290
	inhibit-vs-fix	0.372	0.237	0.211	0.222
	inhibit-vs-shiftupdate	-0.148	-0.176	0.111	0.068
	shift-vs-control	0.128	0.218	0.038	0.172
	shift-vs-fix	0.301	0.494	0.100	0.118
	shift-vs-inhibitupdate	0.231	0.359	0.071	0.123
	update-vs-control	0.179	0.031	-0.074	0.298
	update-vs-fix	0.545	0.537	0.342	0.442
	update-vs-shiftinhibit	-0.036	-0.122	-0.109	0.301
COUNTSW	highlow-vs-fix	0.824	0.473	0.532	0.340
	high-vs-fix	0.787	0.496	0.511	0.322
	high-vs-low	0.340	0.301	0.107	0.106
	low-vs-fix	0.785	0.427	0.503	0.320

Table 2 Continued

Tasks	Contrasts	Within-Subject Correlation		Between-Subject Correlation	
		P1S1-P1S2	P2S1-P2S2	P1S1-P2S1	P1S2-P2S2
SRC	sameopposite-vs-fix	0.631	0.147	0.218	0.118
	opposite-vs-fix	0.554	0.172	0.182	0.100
	opposite-vs-same	-0.017	0.119	0.083	0.067
	same-vs-fix	0.370	-0.029	0.223	0.106
SPAWM	highload-vs-fix	-0.101	0.485	0.129	0.114
	highload-vs-lowload	-0.212	0.549	0.066	-0.164
	highlow-vs-fix	0.265	0.364	0.219	0.296
	lowload-vs-fix	0.113	0.501	0.166	-0.110
OVERT	central-vs-fix	0.357	0.145	-0.227	0.112
	periphcentral-vs-fix	0.594	0.177	0.089	0.429
	peripheral-vs-central	0.353	0.226	0.163	0.026
	peripheral-vs-fix	0.649	0.226	0.361	0.467
COVERT	central-vs-fix	0.297	0.238	0.018	0.118
	periphcentral-vs-fix	0.489	0.344	0.043	0.406
	peripheral-vs-central	0.496	0.549	0.312	0.141
	peripheral-vs-fix	0.575	0.446	0.116	0.332
RNG	baseline-vs-fix	0.268	-0.227	-0.164	0.073
	randbaseline-vs-fix	0.242	0.243	-0.110	-0.043
	random-vs-baseline	0.463	0.091	0.117	0.152
	random-vs-fix	0.508	0.400	0.140	-0.035
GNG	gonogo-vs-fix	0.012	0.221	-0.384	-0.131
	go-vs-fix	-0.032	0.194	-0.363	-0.106
	nogo-vs-fix	0.054	0.252	-0.395	-0.154
	nogo-vs-go	0.039	0.447	0.254	-0.121
ITEMREC	ctrl-vs-fix	0.287	0.401	-0.251	-0.163
	itemrecctrl-vs-fix	0.527	0.300	-0.060	-0.087
	itemrec-vs-ctrl	0.215	0.504	-0.216	-0.133
	itemrec-vs-fix	0.473	0.302	0.000	-0.025

Note. Within- and between-subject similarity calculated from contrast data. Within-subject similarity is greater than between-subject similarity for most contrasts.

Figure 1

Brain Maps with Contrast Data in Right PFC

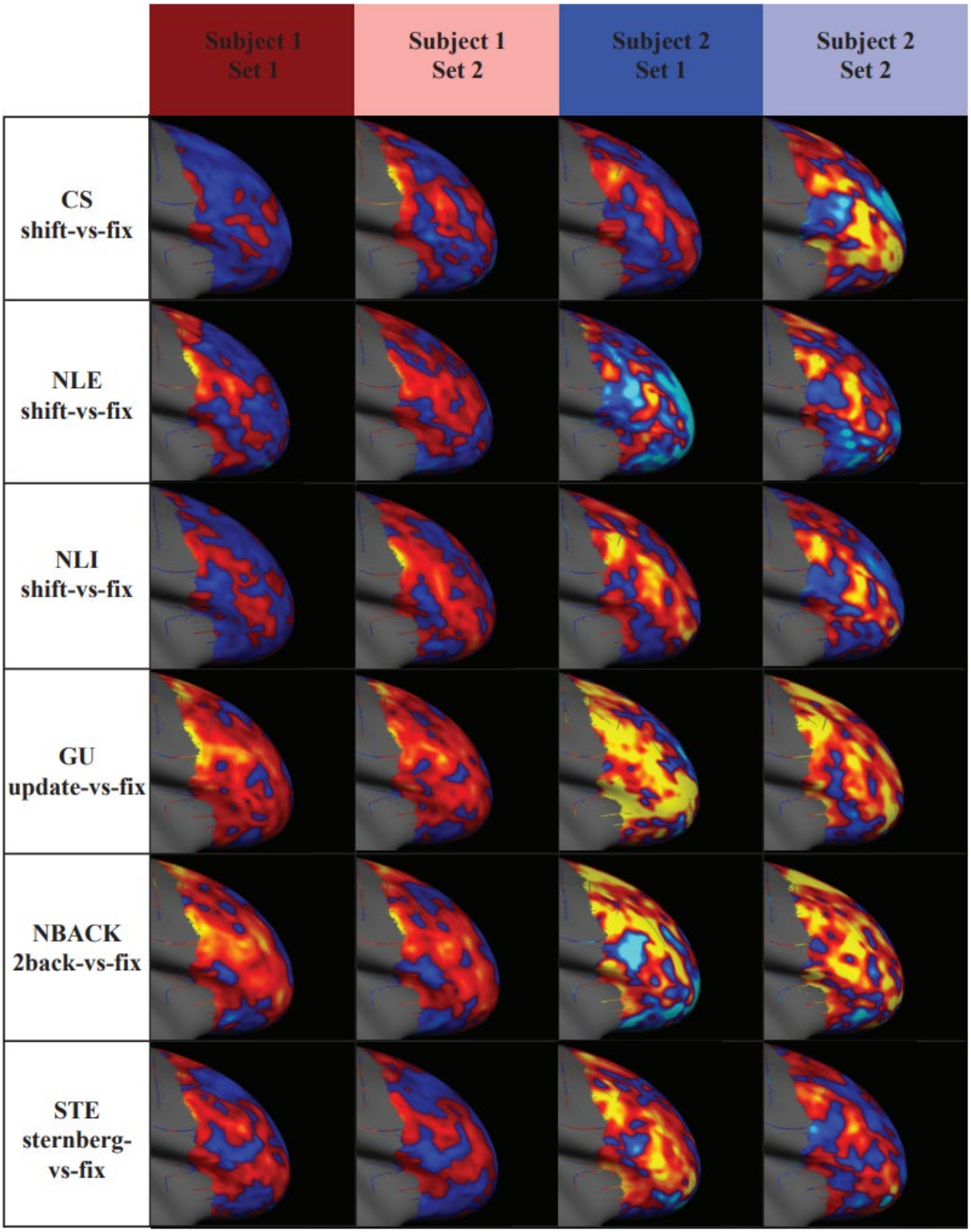


Figure 2 Continued

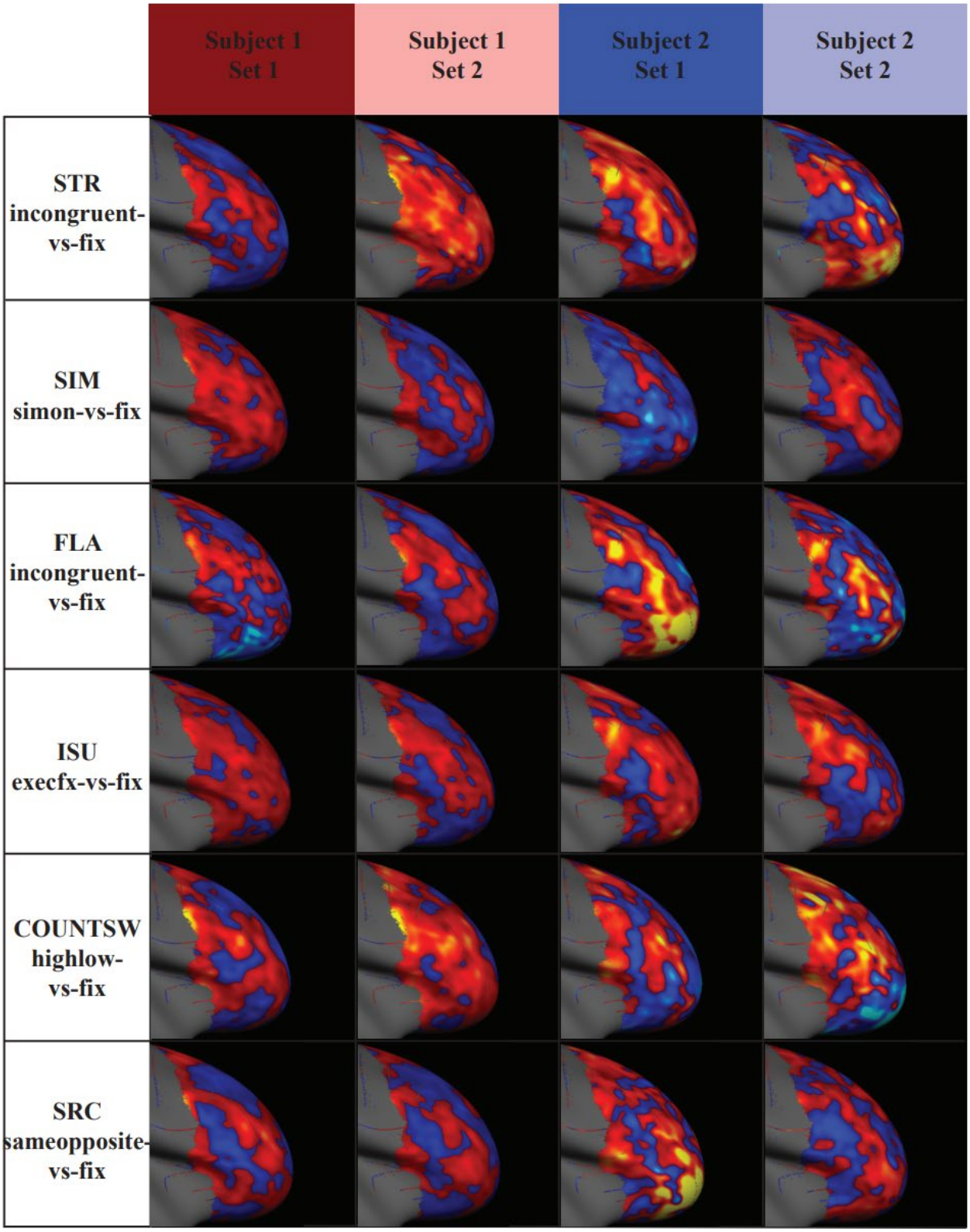
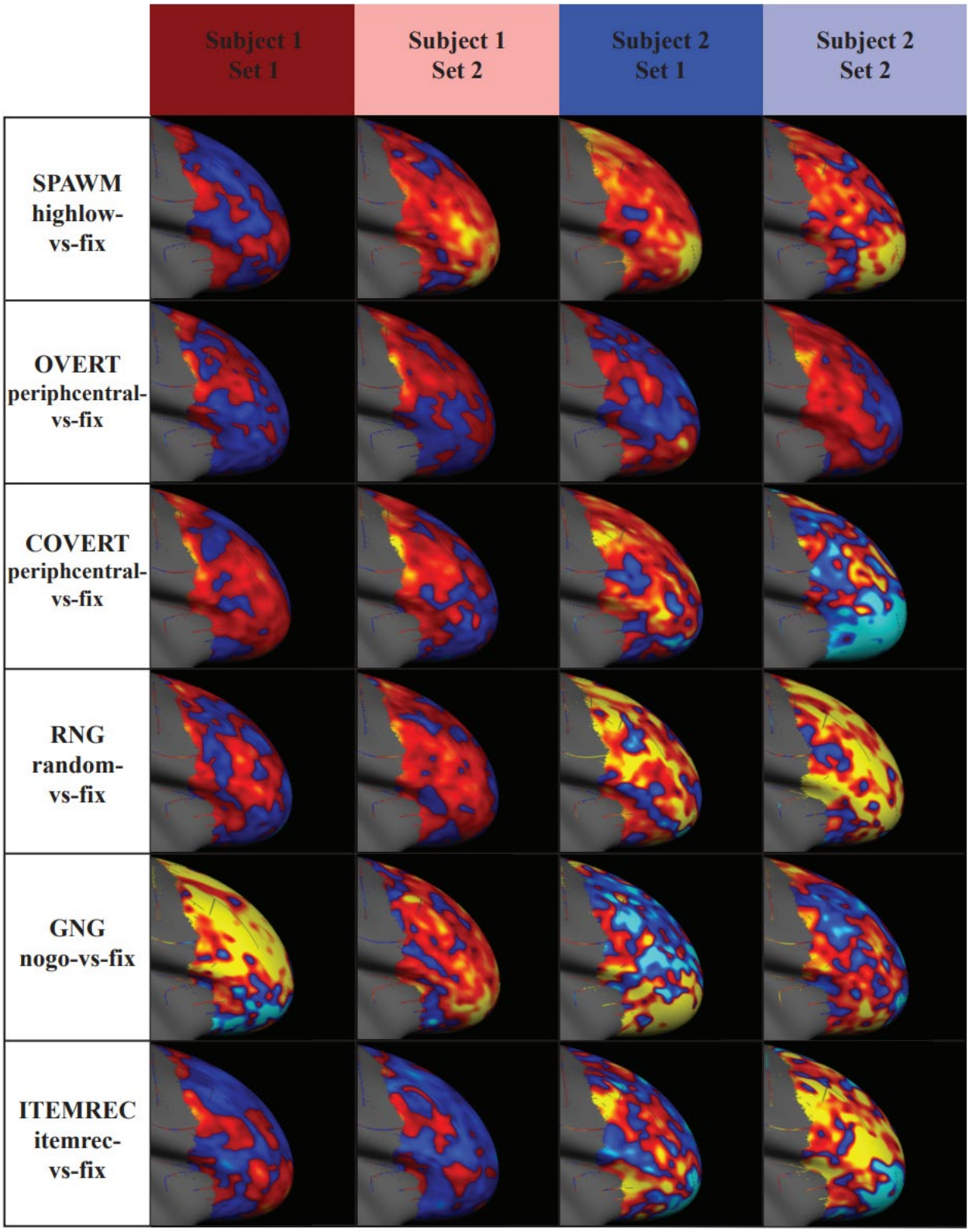


Figure 2 Continued



Note. Brain maps created from contrast data for individual participants and sets. Maps from the same person appear more similar than maps from different people.

Exploratory Factor Analysis

The results of the EFA are summarized in Table 3. The first two factors loaded on contrasts that consisted of a task condition versus resting fixation or a task condition versus an appropriate control condition. Thus, we tentatively identify factor one and factor two as general factors that reflect the fact that the participant is performing a task. The remaining factors loaded separately on different tasks completed in the scanner. Specifically, factor three and factor seven differentiated the color shape task and the go no-go task. Contrasts from inhibit, switch, and update task had high loadings for factor five, and factor eight was associated with contrasts from the number letter task. However, contrasts from the number list task (NLI) were split across two separate factors. NLI contrasts that consisted of a task condition versus resting fixation had high loadings for factor four, while those that consisted of a task condition versus a control condition had high loadings for factor six. We therefore tentatively identified each of these factors as factors for the associated task, except in the case of factor four and six, which we identified as NLI-rest and NLI-shift, respectively. The brain maps generated from these factor scores are shown in Figure 2. In both, within-subject similarity appears consistently higher than between-subject similarity.

Table 3

Factor Loadings of Contrasts for Eight-Factor Solution

Task	Contrast	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6	Factor 7	Factor 8
CS	colorshape-vs-fix			-0.6563					
	color-vs-fix			-0.7273					
	shape-vs-fix								
	shift-vs-color			**0.8520**					
	shift-vs-colorshape			**0.9481**					
	shift-vs-fix		0.5541						
	shift-vs-shape			0.6755					
NLE	letter-vs-fix								0.6992
	numberletter-vs-fix								**0.9644**
	number-vs-fix								0.7484
	shift-vs-fix	0.7074							
	shift-vs-letter	0.4489							
	shift-vs-number	0.6511							
	shift-vs-numberletter	0.6655							
NLI	length-vs-fix				**0.8368**				
	numberlist-vs-fix				**0.8904**				
	shift-vs-fix				0.7264				
	shift-vs-length						0.6896		
	shift-vs-numberlist						**0.9514**		
	shift-vs-value						**0.8843**		
	value-vs-fix				0.6621				
GU	control-vs-fix	0.3146							
	fix-vs-gridcontrol	-0.6007							
	update-vs-control	0.3378							
	update-vs-fix	0.5838							

Table 3 Continued

Task	Contrast	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6	Factor 7	Factor 8
NBACK	0back-vs-fix	0.4068							
	2back-vs-0back	0.4737							
	2back-vs-fix	**0.7511**							
	fix-vs-2back0back	**-0.7589**							
STE	control-vs-fix								
	fix-vs-sternbergcontrol	-0.4576							
	sternberg-vs-control	0.5379							
	sternberg-vs-fix	0.6303							
STR	congruent-vs-fix		0.3054						
	fix-vs-inconcon		-0.4869						
	incongruent-vs-congruent								
	incongruent-vs-fix		0.5352						
SIM	baseline-vs-fix								
	fix-vs-simonbaseline								
	simon-vs-baseline								
	simon-vs-fix		0.3276						
FLA	congruent-vs-fix	0.3174							
	fix-vs-inconcon		-0.3978						
	incongruent-vs-congruent								
	incongruent-vs-fix		0.4319						
ISU	execfx-vs-control					**0.9634**			
	execfx-vs-fix	0.4410							
	inhibit-vs-control					0.7317			
	inhibit-vs-fix	0.3764							
	inhibit-vs-shiftupdate								
	shift-vs-control					**0.8511**			

Table 3 Continued

Task	Contrast	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6	Factor 7	Factor 8
ISU	shift-vs-fix								
	shift-vs-inhibitupdate	-0.4791							
	update-vs-control					0.7449			
	update-vs-fix	0.5889							
	update-vs-shiftinhibit	0.4042							
COUNTSW	highlow-vs-fix	**0.7708**							
	high-vs-fix	0.7351							
	high-vs-low								
	low-vs-fix	0.7484							
SRC	sameopposite-vs-fix		0.6765						
	opposite-vs-fix		0.6847						
	opposite-vs-same		0.4188						
	same-vs-fix		0.4021						
SPAWM	highload-vs-fix		0.4327						
	highload-vs-lowload		0.3933						
	highlow-vs-fix								
	lowload-vs-fix	0.4134							
OVERT	central-vs-fix								
	periphcentral-vs-fix	0.3865							
	peripheral-vs-central	0.3765							
	peripheral-vs-fix	0.4914							
COVERT	central-vs-fix		0.4788						
	periphcentral-vs-fix	0.6543							
	peripheral-vs-central	0.6651							
	peripheral-vs-fix	0.7096							

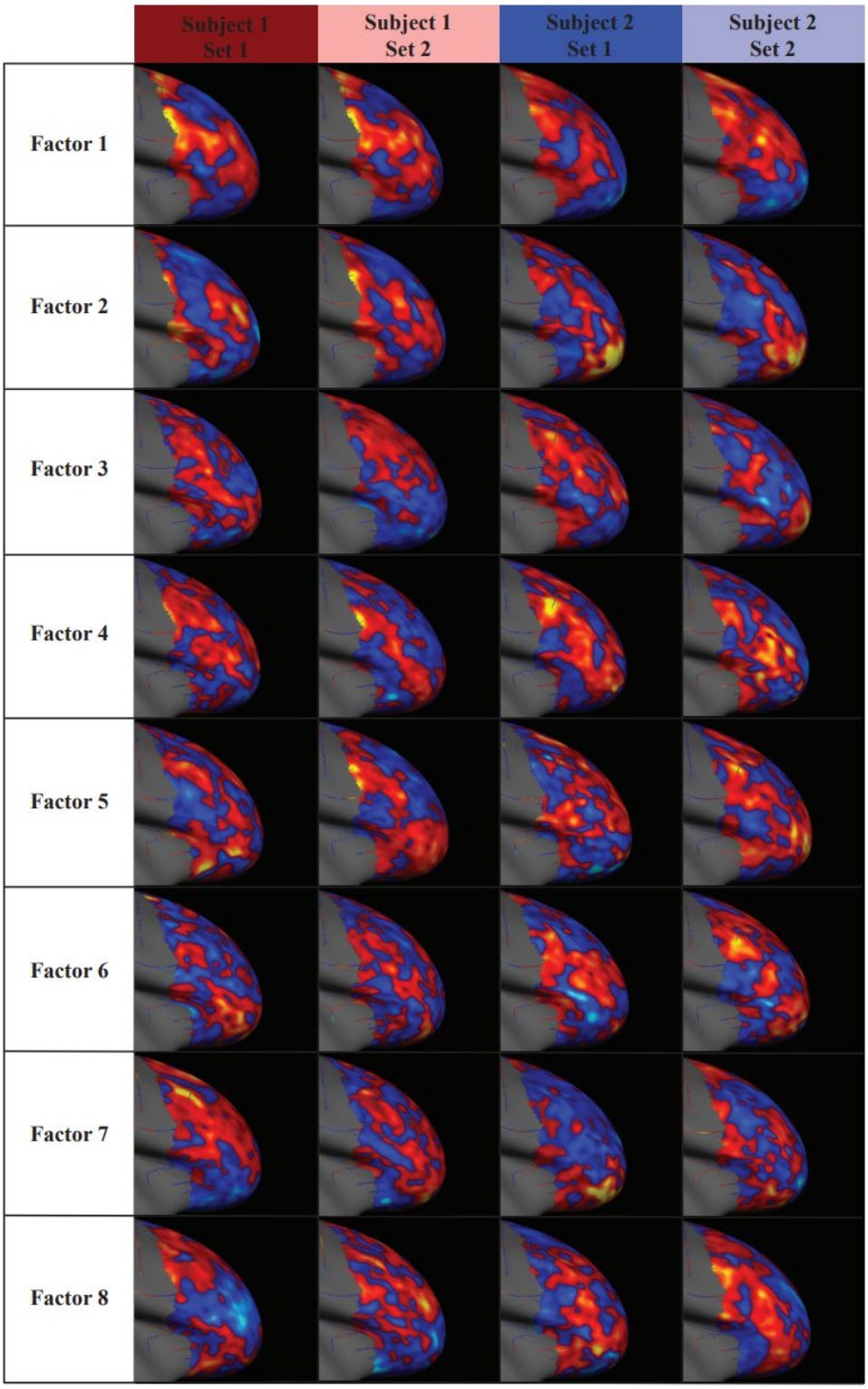
Table 3 Continued

Task	Contrast	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6	Factor 7	Factor 8
RNG	baseline-vs-fix	-0.3150							
	randbaseline-vs-fix		0.4349						
	random-vs-baseline	0.4122							
	random-vs-fix		0.4560						
GNG	gonogo-vs-fix							**0.9829**	
	go-vs-fix							**0.9692**	
	nogo-vs-fix							**0.9784**	
	nogo-vs-go								
ITEMREC	ctrl-vs-fix		0.4127						
	itemrecctrl-vs-fix	0.3538							
	itemrec-vs-ctrl	0.3873							
	itemrec-vs-fix	0.4499							

Note. Factor loadings of the chosen factor solution for each contrast. Loadings less than |0.3| are not shown. Loadings greater than |0.75| are bolded.

Figure 2

Brain Maps with Factor Scores in Right PFC



Note. Brain maps created from factor scores for individual participants and sets. Maps from the same person appear more similar than maps from different people.

Discussion

The present study aimed to address the question of whether the functional organization of a focused region in the right hemisphere, namely the right prefrontal cortex (PFC), is person-specific and reliable. We found that the within-subject similarity of patterns of activation from a large number of cognitive control-related tasks was significantly higher than between-subject similarity. This finding was consistent across almost all 88 contrasts, suggesting that the functional organization of cognitive control in the right PFC is unique to a given individual, but differs from the functional organization of other people. Furthermore, our examination of the functional organization of cognitive control revealed that variations in right PFC activity could be explained by eight latent constructs, which we tentatively associated with performing a task in general and different task sets. Our results provide supporting evidence that using this approach to understand brain function could potentially yield insights that traditional, group-averaged approaches might miss.

This study contributes to the relatively new field of precision neuroimaging and how this new approach can be applied to the study of cognitive control. In particular, our results shed light on how this method can be used to take a closer look at the functional organization of brain areas that are not yet well understood. In previous literature, the functional organization of the right PFC and its relationship with cognitive control processes was not as extensively studied as its left hemisphere counterpart (Goel et al., 2013; Henson et al., 1999). For example, Aron et al. (2014) suggested that the right inferior frontal gyrus is responsible for response inhibition, but that claim has been debated by several others (Erika-Florence et al., 2014; Swick & Chatham, 2014). Yet, this lack of clarity might be due to the individual differences in the patterns of brain activation in the inferior frontal gyrus during inhibitory control (Garavan et al., 2006). Our initial

results demonstrate how the highly reliable measurements obtained through precision neuroimaging can be leveraged to better understand brain function. Specifically, our results demonstrate that functional organization differs across individuals and that traditional group-averaged approaches may be misleading, which may provide an alternative explanation for these conflicts in the literature (Braga & Buckner, 2017; Gordon et al., 2017b; Laumann et al., 2015).

Additionally, the current study presents a new, task-related precision neuroimaging dataset. Though much of the precision neuroimaging literature has focused on resting-state fMRI (RS-fMRI), recent work has proposed that using tasks in precision neuroimaging could yield a clearer picture of individual differences and avoid the potential concern of the psychologically unconstrained nature of the resting state (Greene et al., 2018; Naselaris et al., 2021; Shehzad et al., 2009). Several recent projects have started to incorporate task states into the imaging process. For example, Individual Brain Charting (IBC; Pinho et al., 2018) collected high-resolution task-fMRI data to map a wide range of cognition in the brain (Pinho et al., 2021). Other datasets like Natural Scenes Dataset (NSD; Allen et al., 2022) provided a massive sampling of brain activity in eight individuals as they viewed over 9000 natural scenes. These datasets, like ours, collected data from a few individuals in many different brain states to understand the brain in more detail. Using additional tasks in precision neuroimaging allows us to learn about the neural organization of brain functions at a fine-grained scale (Thirion et al., 2021).

In addition, the factor solution obtained from the EFA is consistent with ongoing work in our lab that will allow us to potentially distinguish between theories of cognitive control. We tentatively associated the first two factors with general factors that reflect the fact that the participant is performing a task, which is consistent with previous factor analyses of executive function (Friedman & Miyake, 2017; Miyake et al., 2000). However, previous factor analyses

have differentiated between cognitive control processes, like inhibition or updating, whereas we associated factors three through eight with the tasks completed during scanning (Miyake et al., 2000). In fact, it seems that in a given individual, many of the same areas in right PFC are the most active across many different task contexts (Figure 1). If this is true, our results might provide evidence for adaptive coding theories. Adaptive coding suggests that neurons become tuned to task-related goals and allow areas such as PFC to be involved in many different cognitive control domains under different contexts (Duncan, 2001). This theory has some support in previous literature. For example, Freedman et al., (2001) tracked the activities of neurons in PFC of primates and found that the activity of individual neurons is highly adaptable to the current stimuli. A previous study by Hampshire et al. (2010) also proposed that the right inferior frontal gyrus is not solely responsible for response inhibition, but also plays a role in maintaining attentional control across multiple task contexts. However, other theories, such as Miyake et al.'s (2000) classic latent variable analysis of executive function, suggest that cognitive control is split up based on processes. In the future, our lab could expand upon this idea by attempting to rigorously test whether the patterns of brain activation we see in these individuals might be able to provide support for any of these theories or assist in creating a new one.

However, this study is not without limitations. One potential concern is that the activation we see in some contrasts may actually be noise, and therefore not attributable to cognitive control processes. In fact, it's possible that some of the contrasts did not produce any significant activation in the right PFC in one or both individuals. Yet, we used the raw contrast effect size (CES) data in our analyses and measured patterns of activation regardless of whether one condition in a given contrast produced significant activation over another. If this was significant

enough to influence the data analysis, it would have likely reduced the within-subject similarity (reliability) of activation. Despite this potential presence of noise in the data, the activations were still consistent across sessions and within individuals, and there was significant evidence indicating that within-person similarity was higher than between-person similarity.

Another potential question concerns whether the difference in patterns of activation across people might be due to differences in brain structure, or even the use of different strategies to complete the tasks. For example, Miller and Van Horn (2007) suggest that some individuals may display patterns of activity that differ from the group average because they use a different strategy, and that such differences could help us better understand individual differences in behavior. Though we did not collect data on strategy use in our participants, we believe this is unlikely due to the tasks used in the study. Our tasks were carefully selected based on previous research, and for the most part, can only be successfully completed using a specific strategy. Moreover, our participants were given specific instructions before the tasks in the scanner, as well as a chance to familiarize themselves with those instructions before scanning. Despite what we believe is limited opportunity for strategic differences in our participants, it is still worthwhile to explore the source of variability in the functional organization among different individuals. Another possible direction for future work may be to collect data on each participant's strategy for completing the tasks, and control for that using statistical methods.

Perhaps one of the most significant limitations of the present study is that we only examined the functional organization of cognitive control in two participants. With this small sample size, we are unable to explore brain-behavior relationships, which is essential to furthering our understanding of cognitive control. A future direction for our lab is to collect more data from more participants, but this issue also begs the question of generalizability. It is

important to know whether these results and precision neuroimaging can tell us anything about the population at large. Similar concerns about the replicability of individual difference research have been raised before, for example, in Waller et al.'s (2017) commentary on connectome fingerprinting, where replications of a classic study by Finn et al. (2015) have obtained different results even when using the same dataset (the Human Connectome Project (HCP); Van Essen et al., 2013). However, we believe that this approach, and our results, still have the potential to improve our understanding of the neural architecture of cognitive control. Firstly, despite these individual differences, all brains still have a lot in common. Secondly, the fine-grained detail provided by the precision neuroimaging approach has allowed for the discovery of details of brain organization that were not visible using traditional, group-averaged approaches (Braga & Buckner, 2017; Braga et al., 2020; Gordon et al., 2020; Gordon & Nelson, 2021; Michon et al., 2022). Thus, looking at even a few people with fine-grained detail is still likely to contribute to our understanding of cognitive control in all people. However, that does not mean that generalizability is not a topic of importance. Generalizability to the larger population has not been extensively studied in the precision neuroimaging literature so far, perhaps due to the use of a small pool of datasets. Importantly, the individuals in these datasets likely don't accurately represent the general population (Michon et al., 2022). Therefore, future work should aim to address both the generalizability of functional organization in the right PFC and the precision neuroimaging approach more broadly.

In conclusion, this study presents precision neuroimaging as a valid and insightful method to study cognitive control and supports a person-specific and reliable functional organization of cognitive control in the right PFC. However, this is only a preliminary step in the investigation of individual differences in the neural representations of cognitive control. We have

only demonstrated that there are indeed individual differences. Future studies should elaborate on this finding, investigating how these differences manifest in different groups of individuals (e.g., populations with psychiatric and neurological disorders) and incorporating generalizability and diversity into consideration. As an ongoing project in our lab, we hope to be able to use this dataset and these results as an initial exploration of the promise of precision neuroimaging and its ability to shed light on the neural underpinnings of cognitive control.

References

Allen, E. J., St-Yves, G., Wu, Y. H., Breedlove, J. L., Prince, J. S., Dowdle, L. T., . . . Kay, K. (2022). A massive 7T fMRI dataset to bridge cognitive neuroscience and artificial intelligence [Article]. *Nature Neuroscience*, *25*(1), 116-+. https://doi.org/10.1038/s41593-021-00962-x

Altamura, M., Elvevag, B., Blasi, G., Bertolino, A., Callicott, J. H., Weinberger, D. R., . . . Goldberg, T. E. (2007). Dissociating the effects of Sternberg working memory demands in prefrontal cortex [Article]. *Psychiatry Research-Neuroimaging*, *154*(2), 103-114. https://doi.org/10.1016/j.pscychresns.2006.08.002

Anderson, J. S., Ferguson, M. A., Lopez-Larson, M., & Yurgelun-Todd, D. (2011). Reproducibility of Single-Subject Functional Connectivity Measurements. *American Journal of Neuroradiology*, *32*(3), 548-555. https://doi.org/10.3174/ajnr.A2330

Aron, A. R., & Poldrack, R. A. (2006). Cortical and subcortical contributions to stop signal response inhibition: Role of the subthalamic nucleus [Article]. *Journal of Neuroscience*, *26*(9), 2424-2433. https://doi.org/10.1523/jneurosci.4682-05.2006

Aron, A. R., Robbins, T. W., & Poldrack, R. A. (2014). inhibitrion and the right inferior frontal cortex: one decade on [Review]. *Trends in Cognitive Sciences*, *18*(4), 177-185. https://doi.org/10.1016/j.tics.2013.12.003

Awh, E., Jonides, J., Smith, E. E., Schumacher, E. H., Koeppe, R. A., & Katz, S. (1996). Dissociation of storage and rehearsal in verbal working memory: Evidence from positron emission tomography [Article]. *Psychological Science*, *7*(1), 25-31. https://doi.org/10.1111/j.1467-9280.1996.tb00662.x

Banich, M. T., Milham, M. P., Atchley, R., Cohen, N. J., Webb, A., Wszalek, T., . . . Magin, R. (2000). fMRI Studies of Stroop Tasks Reveal Unique Roles of Anterior and Posterior Brain Systems in Attentional Selection. *Journal of Cognitive Neuroscience*, *12*(6), 988-1000. https://doi.org/10.1162/08989290051137521

Barbey, A. K., Koenigs, M., & Grafman, J. (2013). Dorsolateral prefrontal contributions to human working memory [Article]. *Cortex*, *49*(5), 1195-1205. https://doi.org/10.1016/j.cortex.2012.05.022

Braga, R. M., & Buckner, R. L. (2017). Parallel Interdigitated Distributed Networks within the Individual Estimated by Intrinsic Functional Connectivity. *Neuron*, *95*(2), 457-+. https://doi.org/10.1016/j.neuron.2017.06.038

Braga, R. M., DiNicola, L. M., Becker, H. C., & Buckner, R. L. (2020). Situating the left-lateralized language network in the broader organization of multiple specialized large-scale distributed networks [Article]. *Journal of Neurophysiology*, *124*(5), 1415-1448. https://doi.org/10.1152/jn.00753.2019

de Haan, B., Moryan, P. S., & Rorden, C. (2008). Covert orienting of attention and overt eye movements activate identical brain regions [Article]. *Brain Research*, *1204*, 102-111. https://doi.org/10.1016/j.brainres.2008.01.105

Demetriou, L., Kowalczyk, O. S., Tyson, G., Bello, T., Newbould, R. D., & Wall, M. B. (2018). A comprehensive evaluation of increasing temporal resolution with multiband-accelerated protocols and effects on statistical outcome measures in fMRI [Article]. *Neuroimage*, *176*, 404-416. https://doi.org/10.1016/j.neuroimage.2018.05.011

Desikan, R. S., Segonne, F., Fischl, B., Quinn, B. T., Dickerson, B. C., Blacker, D., . . . Killiany, R. J. (2006). An automated labeling system for subdividing the human cerebral cortex on

MRI scans into gyral based regions of interest [Article]. *Neuroimage*, *31*(3), 968-980. https://doi.org/10.1016/j.neuroimage.2006.01.021

Desimone, R., & Duncan, J. (1995). Neural Mechanisms of Selective Visual Attention. *Annual Review of Neuroscience*, *18*(1), 193-222. https://doi.org/10.1146/annurev.ne.18.030195.001205

DiGirolamo, G. J., Kramer, A. F., Barad, V., Cepeda, N. J., Weissman, D. H., Milham, M. P., . . . McAuley, E. (2001). General and task-specific frontal lobe recruitment in older adults during executive processes: A fMRI investigation of task-switching [Article]. *Neuroreport*, *12*(9), 2065-2071. https://doi.org/10.1097/00001756-200107030-00054

Dixon, M. L., & Christoff, K. (2012). The Decision to Engage Cognitive Control Is Driven by Expected Reward-Value: Neural and Behavioral Evidence [Article]. *Plos One*, *7*(12), 12, Article e51637. https://doi.org/10.1371/journal.pone.0051637

Dubois, J., & Adolphs, R. (2016). Building a Science of Individual Differences from fMRI. *Trends in Cognitive Sciences*, *20*(6), 425-443. https://doi.org/10.1016/j.tics.2016.03.014

Duncan, J. (2001). An adaptive coding model of neural function in prefrontal cortex. *Nature Reviews Neuroscience*, *2*(11), 820-829. https://doi.org/10.1038/35097575

Erika-Florence, M., Leech, R., & Hampshire, A. (2014). A functional network perspective on response inhibition and attentional control [Article]. *Nature Communications*, *5*, 12, Article 4073. https://doi.org/10.1038/ncomms5073

Fedorenko, E., & Kanwisher, N. (2009). Neuroimaging of Language: Why Hasn't a Clearer Picture Emerged? [Article]. *Language and Linguistics Compass*, *3*(4), 27. https://doi.org/10.1111/j.1749-818x.2009.00143.x

Finn, E. S., Scheinost, D., Finn, D. M., Shen, X. L., Papademetris, X., & Constable, R. T. (2017). Can brain state be manipulated to emphasize individual differences in functional connectivity? *Neuroimage*, *160*, 140-151. https://doi.org/10.1016/j.neuroimage.2017.03.064

Finn, E. S., Shen, X. L., Scheinost, D., Rosenberg, M. D., Huang, J., Chun, M. M., . . . Constable, R. T. (2015). Functional connectome fingerprinting: identifying individuals using patterns of brain connectivity. *Nature Neuroscience*, *18*(11), 1664-1671. https://doi.org/10.1038/nn.4135

Fischl, B. (2012). FreeSurfer [Review]. *Neuroimage*, *62*(2), 774-781. https://doi.org/10.1016/j.neuroimage.2012.01.021

Freedman, D. J., Riesenhuber, M., Poggio, T., & Miller, E. K. (2001). Categorical representation of visual stimuli in the primate prefrontal cortex [Article]. *Science*, *291*(5502), 312-316. https://doi.org/10.1126/science.291.5502.312

Friedman, N. P., & Miyake, A. (2017). Unity and diversity of executive functions: Individual differences as a window on cognitive structure [Review]. *Cortex*, *86*, 186-204. https://doi.org/10.1016/j.cortex.2016.04.023

Gabrys, R. L., Tabri, N., Anisman, H., & Matheson, K. (2018). Cognitive Control and Flexibility in the Context of Stress and Depressive Symptoms: The Cognitive Control and Flexibility Questionnaire [Article]. *Frontiers in Psychology*, *9*, 19, Article 2219. https://doi.org/10.3389/fpsyg.2018.02219

Garavan, H., Hester, R., Murphy, K., Fassbender, C., & Kelly, C. (2006). Individual differences in the functional neuroanatomy of inhibitory control [Article]. *Brain Research*, *1105*, 130-142. https://doi.org/10.1016/j.brainres.2006.03.029

Garavan, H., Ross, T. J., & Stein, E. A. (1999). Right hemispheric dominance of inhibitory control: An event-related functional MRI study [Article]. *Proceedings of the National Academy of Sciences of the United States of America*, *96*(14), 8301-8306. https://doi.org/10.1073/pnas.96.14.8301

Geerligs, L., Rubinov, M., Henson, R. N., & Cam, C. A. N. (2015). State and Trait Components of Functional Connectivity: Individual Differences Vary with Mental State. *Journal of Neuroscience*, *35*(41), 13949-13961. https://doi.org/10.1523/jneurosci.1324-15.2015

Georgiou-Karistianis, N., Sritharan, A., Farrow, M., Cunnington, R., Stout, J., Bradshaw, J., . . . Egan, G. (2007). Increased cortical recruitment in Huntington's disease using a Simon task [Article]. *Neuropsychologia*, *45*(8), 1791-1800. https://doi.org/10.1016/j.neuropsychologia.2006.12.023

Gilbert, S. J., Bird, G., Brindley, R., Frith, C. D., & Burgess, P. W. (2008). Atypical recruitment of medial prefrontal cortex in autism spectrum disorders: An fMRI study of two executive function tasks [Article]. *Neuropsychologia*, *46*(9), 2281-2291. https://doi.org/10.1016/j.neuropsychologia.2008.03.025

Goel, V., Vartanian, O., Bartolo, A., Hakim, L., Ferraro, A. M., Isella, V., . . . Nichelli, P. (2013). Lesions to right prefrontal cortex impair real-world planning through premature commitments [Article]. *Neuropsychologia*, *51*(4), 713-724. https://doi.org/10.1016/j.neuropsychologia.2012.11.029

Gordon, E. M., Laumann, T. O., Adeyemo, B., Gilmore, A. W., Nelson, S. M., Dosenbach, N. U. F., & Petersen, S. E. (2017a). Individual-specific features of brain systems identified with resting state functional correlations. *Neuroimage*, *146*, 918-939. https://doi.org/10.1016/j.neuroimage.2016.08.032

Gordon, E. M., Laumann, T. O., Gilmore, A. W., Newbold, D. J., Greene, D. J., Berg, J. J., . . . Dosenbach, N. U. F. (2017b). Precision Functional Mapping of Individual Human Brains. *Neuron*, *95*(4), 791-+. https://doi.org/10.1016/j.neuron.2017.07.011

Gordon, E. M., Laumann, T. O., Marek, S., Raut, R. V., Gratton, C., Newbold, D. J., . . . Nelson, S. M. (2020). Default-mode network streams for coupling to language and control systems [Article]. *Proceedings of the National Academy of Sciences of the United States of America*, *117*(29), 17308-17319. https://doi.org/10.1073/pnas.2005238117

Gordon, E. M., & Nelson, S. M. (2021). Three types of individual variation in brain networks revealed by single-subject functional connectivity analyses [Article]. *Current Opinion in Behavioral Sciences*, *40*, 79-86. https://doi.org/10.1016/j.cobeha.2021.02.014

Gratton, C., Kraus, B. T., Greene, D. J., Gordon, E. M., Laumann, T. O., Nelson, S. M., . . . Petersen, S. E. (2020). Defining Individual -Specific Functional Neuroanatomy for Precision Psychiatry [Review]. *Biological Psychiatry*, *88*(1), 28-39. https://doi.org/10.1016/j.biopsych.2019.10.026

Gratton, C., Laumann, T. O., Nielsen, A. N., Greene, D. J., Gordon, E. M., Gilmore, A. W., . . . Petersen, S. E. (2018). Functional Brain Networks Are Dominated by Stable Group and Individual Factors, Not Cognitive or Daily Variation. *Neuron*, *98*(2), 439-+. https://doi.org/10.1016/j.neuron.2018.03.035

Greene, A. S., Gao, S. Y., Scheinost, D., & Constable, R. T. (2018). Task-induced brain state manipulation improves prediction of individual traits [Article]. *Nature Communications*, *9*, 13, Article 2807. https://doi.org/10.1038/s41467-018-04920-3

Hakun, J. G., Zhu, Z. D., Johnson, N. F., & Gold, B. T. (2015). Evidence for reduced efficiency and successful compensation in older adults during task switching [Article]. *Cortex*, *64*, 352-362. https://doi.org/10.1016/j.cortex.2014.12.006

Hampshire, A., Chamberlain, S. R., Monti, M. M., Duncan, J., & Owen, A. M. (2010). The role of the right inferior frontal gyrus: inhibition and attentional control [Article]. *Neuroimage*, *50*(3), 1313-1319. https://doi.org/10.1016/j.neuroimage.2009.12.109

Henson, R. N. A., Shallice, T., & Dolan, R. J. (1999). Right prefrontal cortex and episodic memory retrieval: a functional MRI test of the monitoring hypothesis [Article]. *Brain*, *122*, 1367-1381. https://doi.org/10.1093/brain/122.7.1367

Juch, H., Zimine, I., Seghier, M. L., Lazeyras, F., & Fasel, J. H. D. (2005). Anatomical variability of the lateral frontal lobe surface: implication for intersubject variability in language neuroimaging [Article]. *Neuroimage*, *24*(2), 504-514. https://doi.org/10.1016/j.neuroimage.2004.08.037

Kane, M. J., & Engle, R. W. (2002). The role of prefrontal cortex in working-memory capacity, executive attention, and general fluid intelligence: An individual-differences perspective [Review]. *Psychonomic Bulletin & Review*, *9*(4), 637-671. https://doi.org/10.3758/bf03196323

Kikuchi, T., Miller, J. M., Schneck, N., Oquendo, M. A., Mann, J. J., Parsey, R. V., & Keilp, J. G. (2012). Neural responses to incongruency in a blocked-trial Stroop fMRI task in major depressive disorder [Article]. *Journal of Affective Disorders*, *143*(1-3), 241-247. https://doi.org/10.1016/j.jad.2012.05.016

Laumann, T. O., Gordon, E. M., Adeyemo, B., Snyder, A. Z., Joo, S. J., Chen, M. Y., . . . Petersen, S. E. (2015). Functional System and Areal Organization of a Highly Sampled

Individual Human Brain. *Neuron*, *87*(3), 657-670. https://doi.org/10.1016/j.neuron.2015.06.037

Lemire-Rodger, S., Lam, J., Viviano, J. D., Stevens, W. D., Spreng, R. N., & Turner, G. R. (2019). Inhibit, switch, and update: A within-subject fMRI investigation of executive control [Article]. *Neuropsychologia*, *132*, 12, Article 107134. https://doi.org/10.1016/j.neuropsychologia.2019.107134

Leung, H. C., Oh, H., Ferri, J., & Yi, Y. J. (2007). Load response functions in the human spatial working memory circuit during location memory updating [Article]. *Neuroimage*, *35*(1), 368-377. https://doi.org/10.1016/j.neuroimage.2006.12.012

Levy, B. J., & Wagner, A. D. (2011). Cognitive control and right ventrolateral prefrontal cortex: reflexive reorienting, motor inhibition, and action updating. In M. B. Miller & A. Kingstone (Eds.), *Year in Cognitive Neuroscience* (Vol. 1224, pp. 40-62). Blackwell Science Publ. https://doi.org/10.1111/j.1749-6632.2011.05958.x

MacDonald, A. W., Cohen, J. D., Stenger, V. A., & Carter, C. S. (2000). Dissociating the role of the dorsolateral prefrontal and anterior cingulate cortex in cognitive control [Article]. *Science*, *288*(5472), 1835-1838. https://doi.org/10.1126/science.288.5472.1835

Menon, V., & D'Esposito, M. (2022). The role of PFC networks in cognitive control and executive function. *Neuropsychopharmacology*, *47*(1), 90-103. https://doi.org/10.1038/s41386-021-01152-w

Michon, K., J., Khammash, D., Simmonite, M., Hamlin, A. M., & Polk, T. A. (2022). Person-specific and precision neuroimaging: Current methods and future directions. *NeuroImage*, *263*, 119589. https://doi.org/https://doi.org/10.1016/j.neuroimage.2022.119589

Miller, E. K., & Cohen, J. D. (2001). An integrative theory of prefrontal cortex function [Review]. *Annual Review of Neuroscience*, *24*, 167-202. https://doi.org/10.1146/annurev.neuro.24.1.167

Miller, M. B., & Van Horn, J. D. (2007). Individual variability in brain activations associated with episodic retrieval: A role for large-scale databases [Article]. *International Journal of Psychophysiology*, *63*(2), 205-213. https://doi.org/10.1016/j.ijpsycho.2006.03.019

Miyake, A., Friedman, N. P., Emerson, M. J., Witzki, A. H., Howerter, A., & Wager, T. D. (2000). The unity and diversity of executive functions and their contributions to complex "frontal lobe" tasks: A latent variable analysis [Article; Proceedings Paper]. *Cognitive Psychology*, *41*(1), 49-100. https://doi.org/10.1006/cogp.1999.0734

Mueller, S., Wang, D. H., Fox, M. D., Yeo, B. T. T., Sepulcre, J., Sabuncu, M. R., . . . Liu, H. S. (2013). Individual Variability in Functional Connectivity Architecture of the Human Brain. *Neuron*, *77*(3), 586-595. https://doi.org/10.1016/j.neuron.2012.12.028

Naselaris, T., Allen, E., & Kay, K. (2021). Extensive sampling for complete models of individual brains [Article]. *Current Opinion in Behavioral Sciences*, *40*, 45-51. https://doi.org/10.1016/j.cobeha.2020.12.008

Peirce, J., Gray, J. R., Simpson, S., MacAskill, M., Hochenberger, R., Sogo, H., . . . Lindelov, J. K. (2019). PsychoPy2: Experiments in behavior made easy [Article]. *Behavior Research Methods*, *51*(1), 195-203. https://doi.org/10.3758/s13428-018-01193-y

Pinho, A. L., Amadon, A., Fabre, M., Dohmatob, E., Denghien, I., Torre, J. J., . . . Thirion, B. (2021). Subject-specific segregation of functional territories based on deep phenotyping [Article]. *Human Brain Mapping*, *42*(4), 841-870. https://doi.org/10.1002/hbm.25189

Pinho, A. L., Amadon, A., Ruest, T., Fabre, M., Dohmatob, E., Denghien, I., . . . Thirion, B. (2018). Individual Brain Charting, a high-resolution fMRI dataset for cognitive mapping. *Scientific Data*, *5*, Article 180105. https://doi.org/10.1038/sdata.2018.105

Reineberg, A. E., Gustavson, D. E., Benca, C., Banich, M. T., & Friedman, N. P. (2018). The Relationship Between Resting State Network Connectivity and Individual Differences in Executive Functions [Article]. *Frontiers in Psychology*, *9*, 14, Article 1600. https://doi.org/10.3389/fpsyg.2018.01600

Ridderinkhof, K. R., Ullsperger, M., Crone, E. A., & Nieuwenhuis, S. (2004). The role of the medial frontal cortex in cognitive control [Review]. *Science*, *306*(5695), 443-447. https://doi.org/10.1126/science.1100301

Salehi, M., Greene, A. S., Karbasi, A., Shen, X. L., Scheinost, D., & Constable, R. T. (2020). There is no single functional atlas even for a single individual: Functional parcel definitions change with task. *Neuroimage*, *208*, Article 116366. https://doi.org/10.1016/j.neuroimage.2019.116366

Seitzman, B. A., Gratton, C., Laumann, T. O., Gordon, E. M., Adeyemo, B., Dworetsky, A., . . . Petersen, S. E. (2019). Trait-like variants in human functional brain networks. *Proceedings of the National Academy of Sciences of the United States of America*, *116*(45), 22851-22861. https://doi.org/10.1073/pnas.1902932116

Shehzad, Z., Kelly, A. M. C., Reiss, P. T., Gee, D. G., Gotimer, K., Uddin, L. Q., . . . Milham, M. P. (2009). The Resting Brain: Unconstrained yet Reliable [Article]. *Cerebral Cortex*, *19*(10), 2209-2229. https://doi.org/10.1093/cercor/bhn256

Silk, T. J., Bellgrove, M. A., Wrafter, P., Mattingley, J. B., & Cunnington, R. (2010). Spatial working memory and spatial attention rely on common neural processes in the

intraparietal sulcus [Article]. *Neuroimage*, *53*(2), 718-724. https://doi.org/10.1016/j.neuroimage.2010.06.068

Smith, M. D., Perez, C. D., Porter, A., Dworetsky, A., & Gratton, C. (2021). Light through the fog: using precision fMRI data to the neural substrates of control [Article]. *Current Opinion in Behavioral Sciences*, *40*, 19-26. https://doi.org/10.1016/j.cobeha.2020.12.004

Suda, A., Osada, T., Ogawa, A., Tanaka, M., Kamagata, K., Aoki, S., . . . Konishi, S. (2020). Functional Organization for Response Inhibition in the Right Inferior Frontal Cortex of Individual Human Brains [Article]. *Cerebral Cortex*, *30*(12), 6325-6335. https://doi.org/10.1093/cercor/bhaa188

Swick, D., & Chatham, C. H. (2014). Ten years of inhibition revisited [Editorial Material]. *Frontiers in Human Neuroscience*, *8*, 2, Article 329. https://doi.org/10.3389/fnhum.2014.00329

Sylvester, C. Y. C., Wager, T. D., Lacey, S. C., Hernandez, L., Nichols, T. E., Smith, E. E., & Jonides, J. (2003). Switching attention and resolving interference: fMRI measures of executive functions [Article]. *Neuropsychologia*, *41*(3), 357-370, Article Pii s0028-3932(02)00167-7. https://doi.org/10.1016/s0028-3932(02)00167-7

Thirion, B., Thual, A., & Pinho, A. L. (2021). From deep brain phenotyping to functional atlasing [Article]. *Current Opinion in Behavioral Sciences*, *40*, 201-212. https://doi.org/10.1016/j.cobeha.2021.05.004

Tomaiuolo, F., MacDonald, J. D., Caramanos, Z., Posner, G., Chiavaras, M., Evans, A. C., & Petrides, M. (1999). Morphology, morphometry and probability mapping of the pars opercularis of the inferior frontal gyrus: an in vivo MRI analysis [Article]. *European*

Journal of Neuroscience, *11*(9), 3033-3046. https://doi.org/10.1046/j.1460-9568.1999.00718.x

Van Essen, D. C., Smith, S. M., Barch, D. M., Behrens, T. E. J., Yacoub, E., Ugurbil, K., & Consortium, W. U.-M. H. (2013). The WU-Minn Human Connectome Project: An overview. *Neuroimage*, *80*, 62-79. https://doi.org/10.1016/j.neuroimage.2013.05.041

Van Horn, J. D., Grafton, S. T., & Miller, M. B. (2008). Individual Variability in Brain Activity: A Nuisance or an Opportunity? *Brain Imaging and Behavior*, *2*(4), 327-334. https://doi.org/10.1007/s11682-008-9049-9

Wager, T. D., Sylvester, C. Y. C., Lacey, S. C., Nee, D. E., Franklin, M., & Jonides, J. (2005). Common and unique components of response inhibition revealed by fMRI [Article]. *Neuroimage*, *27*(2), 323-340. https://doi.org/10.1016/j.neuroimage.2005.01.054

Waller, L., Walter, H., Kruschwitz, J. D., Reuter, L., Muller, S., Erk, S., & Veer, I. M. (2017). Evaluating the replicability, specificity, and generalizability of connectome fingerprints [Article]. *Neuroimage*, *158*, 371-377. https://doi.org/10.1016/j.neuroimage.2017.07.016

Wikenheiser, A. M., & Schoenbaum, G. (2016). Over the river, through the woods: cognitive maps in the hippocampus and orbitofrontal cortex. *Nature Reviews Neuroscience*, *17*(8), 513-523. https://doi.org/10.1038/nrn.2016.56